



BHARATIYA ANTARIKSH HACKATHON 2025

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Team Name : Technocrats

Team Leader Name : Rochan Awasthi

Problem Statement : AI/ML driven automated feature detection and change analysis of glacial lakes from multi-source satellite imagery.

Team Members

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Brief about the Idea:

Glacial lakes are rapidly expanding due to climate change, significantly increasing the risk of **Glacial Lake Outburst Floods (GLOFs)** that threaten downstream communities. Reliable, frequent monitoring is critical for early warning and disaster risk reduction, yet existing methods are often manual, slow, or ineffective in cloud-covered, snow-prone mountainous regions.

Our project addresses this by developing an end to end AI-powered system that automatically detects, segments, classifies, and tracks glacial lakes through the fusion of multiple satellite data sources: **radar imagery** from **Sentinel-1**, **high resolution optical data** from **Sentinel-2** and **Landsat-8/9**, **thermal bands** from **Landsat** to distinguish water from ice or snow, and **Digital Elevation Models (DEM)** for topographic and flood risk analysis.

At the core lies a **hybrid deep learning architecture** that combines **GLNet's multi-scale global-local feature extraction** with **Attention U-Net's spatial attention** and **boundary-aware optimization**. This enables highly accurate lake delineation even under challenging conditions like cloud cover and complex terrain.

- Unlike traditional or single-source approaches, our system's **multi-modal fusion** leverages complementary data strengths, delivering superior accuracy and robustness.
- Additionally, the model incorporates **uncertainty quantification using Monte Carlo Dropout**, producing **pixel-wise confidence maps** that provide trustworthy, risk-aware insights crucial for prioritizing monitoring and issuing early warnings.
- By automating the entire pipeline, from **multi-modal data fusion** and precise **deep learning-based segmentation** to **time series extraction** and **quantitative risk analysis**, our solution delivers a robust, scalable platform for glacial lake monitoring, setting a new standard for disaster preparedness and climate resilience.

How is this different?

- Most existing methods rely on *single-source data* (typically only optical or radar imagery) and standard models like U-Net or thresholding, making them vulnerable to **clouds, shadows, snow, and complex terrain**.
- Our model uniquely fuses radar (**Sentinel-1 SAR**), optical (**Sentinel-2, Landsat-8/9**), thermal (Landsat-8/9), and elevation (DEM) data, leveraging the complementary strengths of each for **robust, all-weather, all-terrain monitoring**.
- The architecture combines **GLNet's multi-scale global-local feature extraction** with **Attention U-Net's spatial focus** and **boundary-aware loss**, enabling **highly accurate segmentation** even in complex environments.
- **Post-processing with superpixels and DenseCRF** further refines boundaries for better detection in challenging terrains.

How does it solve the problem?

- **Multimodal data fusion** overcomes the limitations of single-sensor approaches , ensuring **reliable detection and segmentation** of glacial lakes under any **weather, season, or surface conditions**.
- The **hybrid deep learning model** captures **small lakes** and **complex boundaries** with **high precision**.
- The fully automated pipeline covers everything from data fusion and **segmentation to time series analysis and hazard detection**, delivering actionable insights in near-real time.
- Pixel-wise confidence scores help prioritize high-risk lakes, supporting targeted disaster response and **early warning** for **Glacial Lake Outburst Floods (GLOFs)**

Unique Selling Proposition (USP)

Advanced Hybrid Deep Learning Architecture

- Combines GNet's global-local context learning with Attention U-Net's focus on spatial precision and boundary refinement delivering high segmentation accuracy even in rugged mountain terrains.

Robust Multimodal Data Fusion

- Integrates Sentinel-1 (SAR), Sentinel-2/Landsat (optical), Landsat thermal, and DEM data ensuring reliable detection across all weather, lighting, and topographic conditions.

Pixel-Level Uncertainty Estimation

- Uses Monte Carlo Dropout to generate confidence maps for each pixel supporting risk-aware decision-making and increasing trust in model outputs.

Fully Automated Processing Pipeline

- Automates the entire workflow from raw satellite imagery to glacial lake mapping, trend analysis, and GLOF alerts ready for large-scale operational use.

Enhanced Boundary Accuracy

- Applies SLIC superpixels and DenseCRF in post-processing to sharpen lake edges and reduce false positives improving the overall reliability of the segmentation.

Automated Lake Detection and Mapping

Deep convolutional neural networks to automatically identify and map glacial lakes from multi-source remote sensing data, providing high accuracy and reducing manual effort.

Real-Time Monitoring and Alerts

Continuously monitors glacial lakes and provides real-time alerts to relevant authorities if potential threats or abnormal changes are detected, enabling timely risk mitigation.

Comprehensive Risk Assessment

Integrates multiple parameters such as lake area, water level, temperature, and historical trends to deliver a thorough assessment of glacial lake outburst flood (GLOF) risks



Forecasting and Early Warning

Employs predictive analytics to forecast possible lake expansion or outburst events, offering early warnings that help in disaster preparedness and response planning.

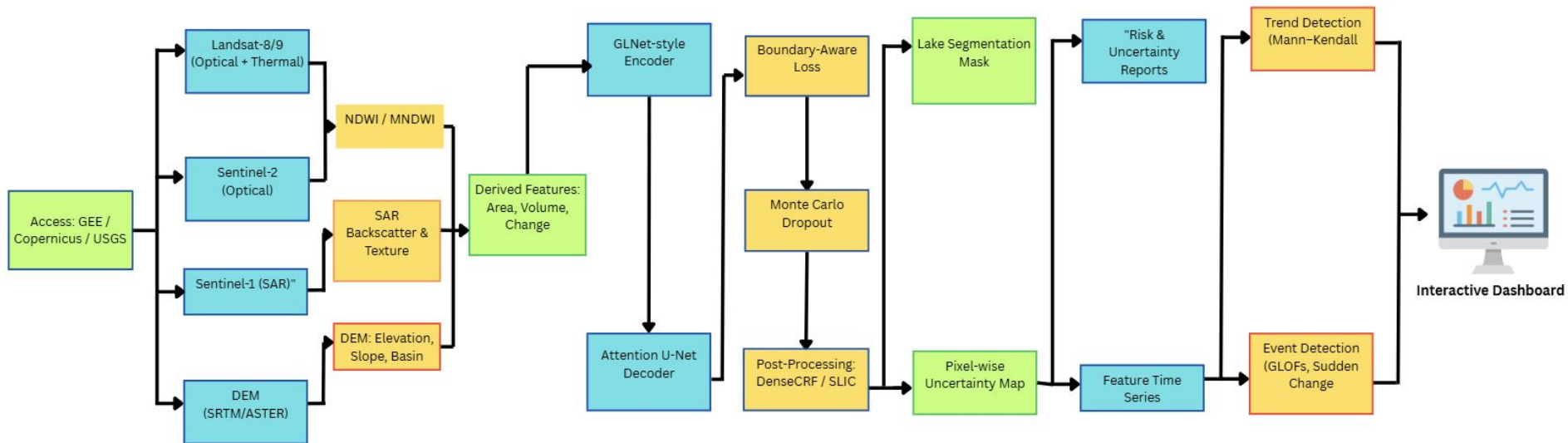
User-Friendly Dashboard

Presents all monitoring results, risk levels, and alerts through an intuitive dashboard accessible via web or mobile, making data and insights easily available to stakeholders and decision-makers.

Scalable and Modular Architecture

Designed to be easily scalable for additional lakes or regions and modular to integrate new data sources, algorithms, or features as technology evolves.

Process Flow Diagram

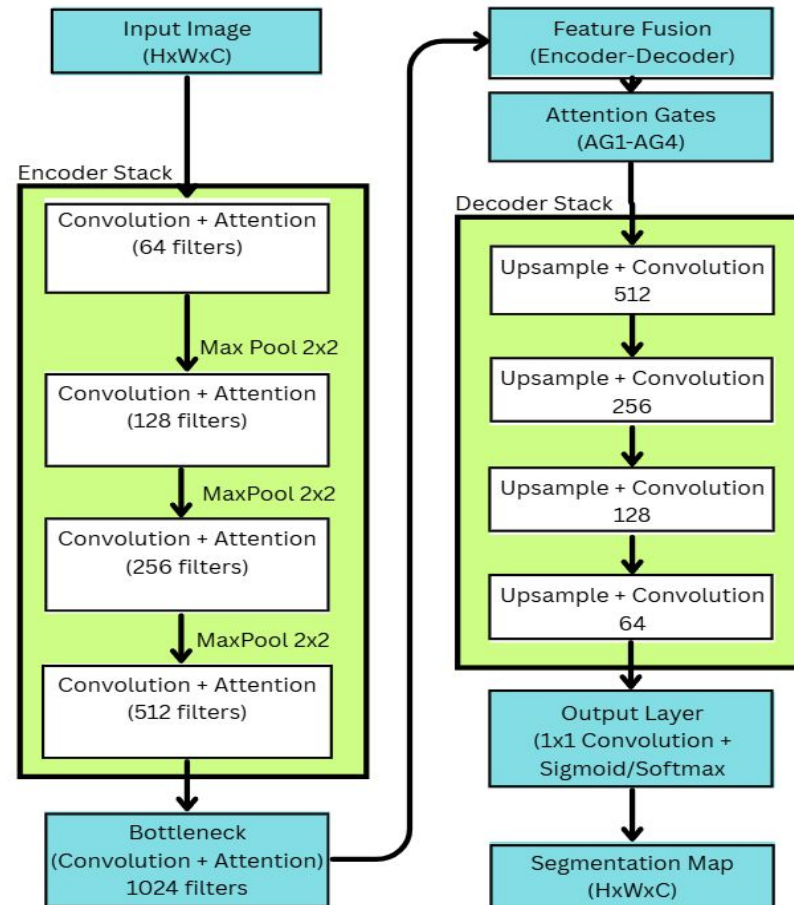


Wireframes/Mock diagrams of the proposed solution (optional)

N/A

Architecture diagram of the proposed solution

- The architecture is based on an enhanced U-Net framework, widely used for glacial lake segmentation.
- It integrates multi-scale global-local feature extraction (GLNet) to capture both broad context and fine details.
- Attention mechanisms (Attention U-Net) are used to focus on important regions and improve segmentation accuracy, especially in complex or noisy areas.
- A boundary-aware loss is applied to sharpen lake boundaries and improve detection of small or debris-covered lakes.
- The model fuses radar, optical, thermal, and elevation data at multiple stages, enabling robust, all-weather segmentation.



Technologies to be used in the solution:



Multi-Modal Data Acquisition

Integrating diverse satellite sources for robust, all-condition monitoring.

- Sentinel-1 SAR (radar imaging)
- Sentinel-2 optical imagery
- Landsat-8/9 optical & thermal
- SRTM/ASTER GDEM (DEM data)
- Google Earth Engine access
- Copernicus Data Space
- USGS EarthExplorer



Advanced Preprocessing & Feature Engineering

Preparing and enriching data for optimal model performance.

- Atmospheric & radiometric correction
- Cloud & snow masking
- NDWI, MNDWI, NDVI indices
- SAR backscatter & texture
- DEM elevation & slope extraction
- Data alignment & resampling
- Feature selection (e.g., PCA)

Hybrid Deep Learning Architecture

State-of-the-art segmentation and multi-modal fusion.

- GLNet multi-scale feature extraction
- Attention U-Net spatial focus
- Boundary-aware loss functions
- Multi-modal input fusion
- Skip connections for detail
- Data augmentation techniques
- PyTorch/TensorFlow frameworks



Uncertainty Quantification & Visualization

Delivering confidence-aware, actionable insights

- Monte Carlo Dropout
- Pixel-wise confidence mapping
- Time series trend analysis
- GLOF risk detection
- Interactive dashboards (Streamlit)
- QGIS, Plotly visualization
- Automated reporting

Estimated implementation cost

To implement a fully automated, pilot-scale glacial lake monitoring system, we require cloud infrastructure for storing satellite data, running AI model training and inference, automating workflows, and hosting a user-friendly dashboard. Leveraging AWS services with cost-saving options like spot instances and archival storage enables an efficient and affordable solution without additional human labour costs.

Key Cost Components (Annual, INR)

- EC2 GPU Compute (₹35,000 – ₹80,000):
Spot instances to handle AI model training and inference efficiently.
- Amazon S3 Storage (₹40,000 – ₹1,00,000):
Combination of S3 Standard for active data and Glacier for archival storage.
- Dashboard Hosting (₹1,000):
Static website hosted on S3 for low-cost, reliable access.
- Workflow Automation (₹10,000 – ₹20,000):
AWS Lambda and Step Functions for orchestrating data processing pipelines.
- Maintenance & Updates (₹10,000 – ₹30,000):
Cloud infrastructure upkeep, backups, and scaling.

Total Estimated Annual Cost:

₹96,000 – ₹2,31,000

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THANK YOU

